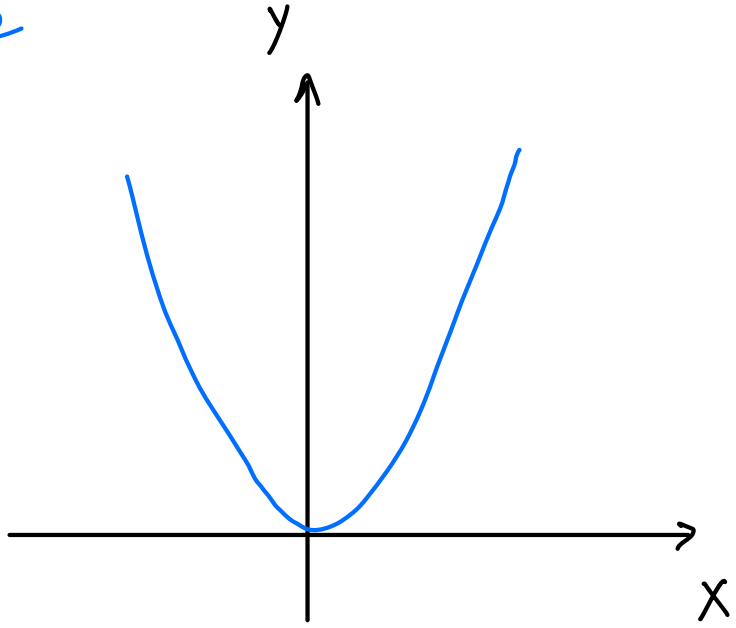
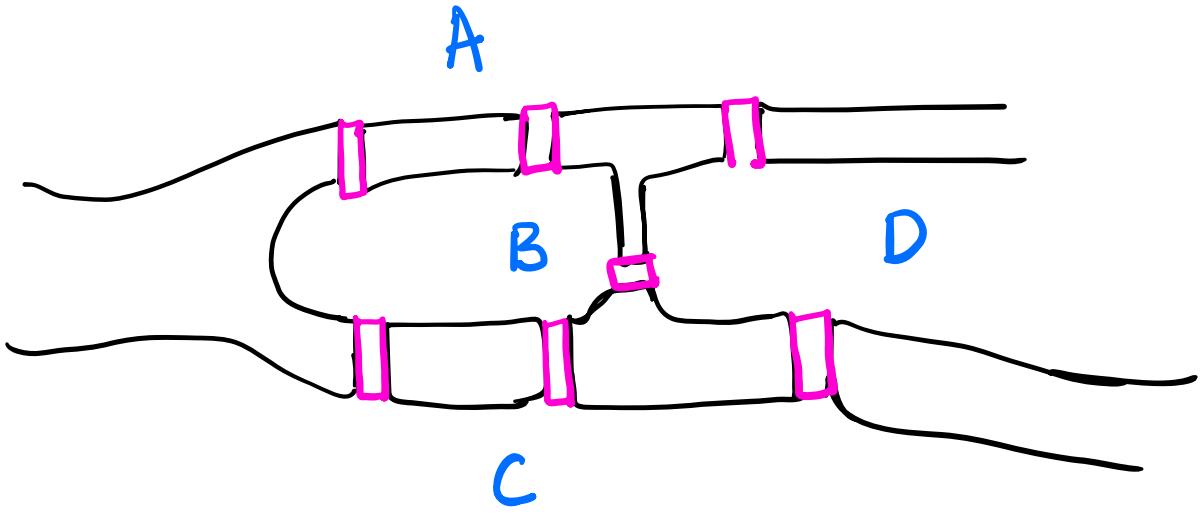
Graphs : The Basics

$y = x^2$



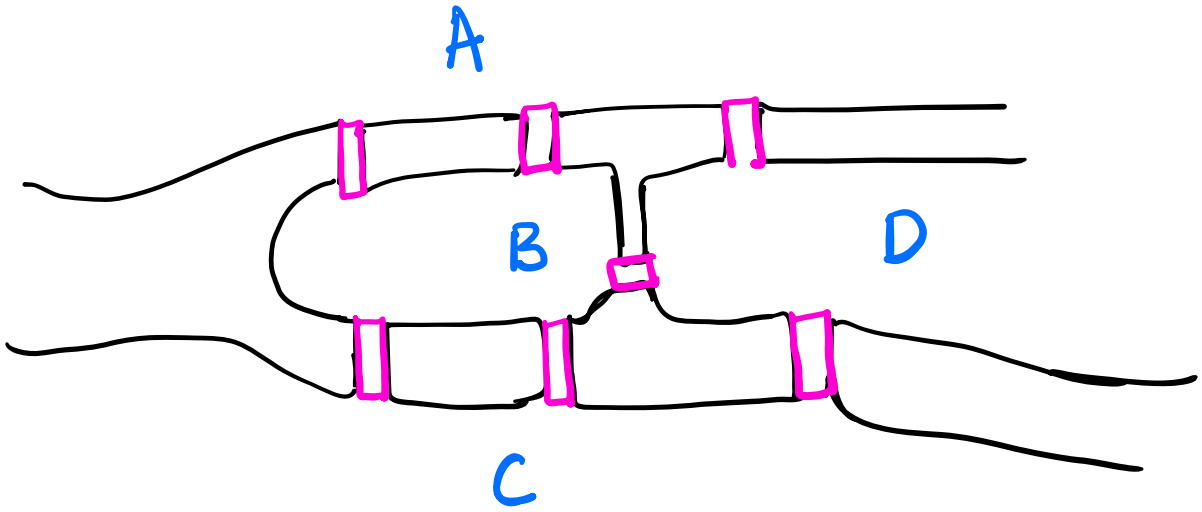
Graph : Not this

Königsburg Bridges Problem



The city of Königsburg exists as collection of islands connected by bridges as shown in figure.

Königsburg Bridges Problem



The city of Königsburg exists as collection of islands connected by bridges as shown in figure.

Q: Is it possible to start at some location in the town, travel across all the bridges once without crossing any bridge twice and return to the starting Point.

Euler in 1736 solved this very first Problem in Graph Theory

the field has exploded becoming one of the most

important areas of Mathematics and Computer Science.

Graphs have applications in many fields

- Social networks
- Road networks
- Scheduling
- etc

Today's Plan

- Recap Basic definitions
- Representation of graph

- A graph is a representation of a set of objects (such as people) and pairwise relationships between them (such as friendships)

- The objects are called the vertices or the nodes of the graph.

- The pairwis relationships translate to the edges of the graph.

Formal Definition

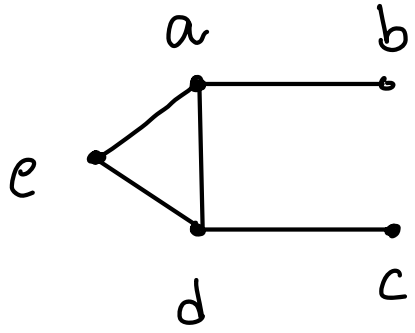
A graph G is a pair of sets (V, E), where V is an arbitrary non-empty set and E is a set of pairs of elements of V.

Elements of V and E are called vertices (nodes) and edges (links) of G respectively.

Examples :

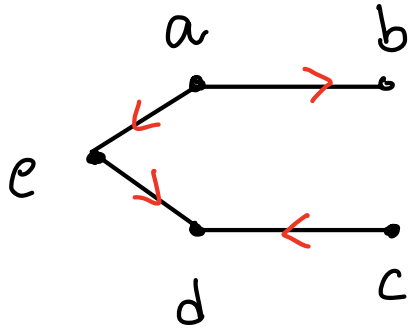
① $V = \{a, b, c, d, e\}$ [Undirected graph]

$E = \{\{a,b\}, \{c,d\}, \{d,e\}, \{e,a\}, \{a,d\}\}$



② $V = \{a, b, c, d, e\}$ [Directed graph]

$E = \{(a,b), (c,d), (e,d), (a,e)\}$



Undirected graphs vs Directed graphs

- Facebook
- Instagram
- Road networks (distance)
- Airport networks (Price)
- Conflict graph [Students]
- Directed conflict graph

Summary Sofar

Graphs can be

- Undirected
- Directed
- Undirected & edge weighted
- Directed & edge weighted
- Undirected & Vertex weighted
- Directed & Vertex weighted

Notation for Graphs

For a graph $G = (V, E)$ with vertex set V and edge set E

* – $n = |V|$ denote the number of vertices
* – $m = |E|$ denotes the number of edges

Size of a graph

Recap: We analyze the running time of different algorithms as a function of the input size.

When the input involves a graph we must specify exactly how the graph is represented and what we mean by its size.

Other Terminology

- Adjacent Vertices
- degree of a Vertex
- Simple graphs
- Path
- Distance in graphs
- Connected graph

} Today (Revision)

I will recap the terminology whenever it is required

Q: How to represent graphs ?

Representation of Graphs

There are several ways to represent graphs, each with its advantages and disadvantages.

In this course we will see three ways to represent graphs.

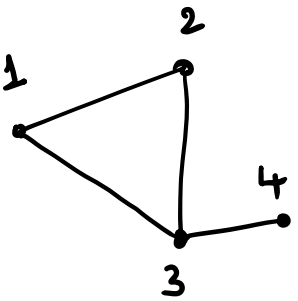
1. Edge lists
2. Adjacency matrices
3. Adjacency Lists

} Most Commonly used

Edge list

An edge list is a list or array of all the edges of the graph.

E.g.:



Array = [[1 2] [2 3] [1 3] [3 4]]

Total space for an edge list is O(E)

disadvantage:

If we want to find whether the graph contains a particular edge, we have to search through the edge list, In the worst case we have to search through $|E|$ edges.

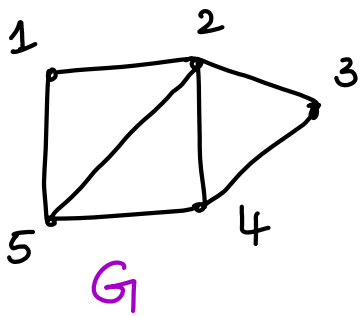
Adjacency matrices

We assume that the vertices are numbered $1, 2, \ldots |V|$ in some arbitrary manner.

The adjacency matrix of a graph G is a $|V| \times |V|$ matrix $A = (a_{ij})$ of 0's and 1's such that

$$a_{ij} = \begin{cases} 1 & \text{if } ij \in E \\ 0 & \text{otherwise} \end{cases}$$

Example:



G

$$A = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Adjacency matrix representation of G.

Q: How much space does the adjacency matrix of a graph require, as a function of the number n of vertices and the number m of edges?

Remarks:

① For an undirected graph, the adjacency matrix is Symmetric.

For a directed graph the adjacency matrix need not be Symmetric.

② The adjacency matrix of a graph depends on the ordering of the Vertices.

Advantages

We can find out whether an edge is present in constant time by just looking up the corresponding entry in the matrix

Disadvantages

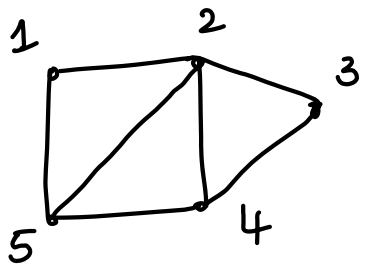
① Adjacency matrix representation of a graph requires $\Theta(n^2)$ space, irrespective of the # of edges in the graph.

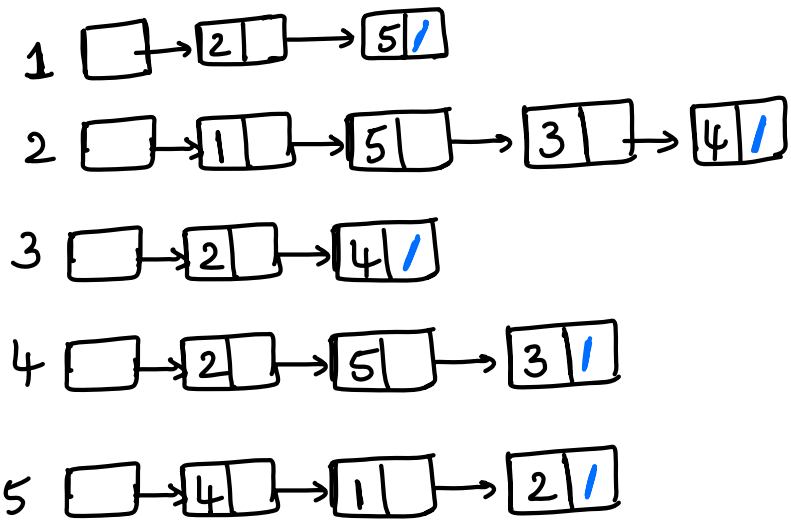
② Finding which vertices are adjacent to a given vertex i, we have to look at all $|V|$ entries in row i, even if only a small number of vertices are adjacent to a vertex i

Adjacency Lists:

An adjacency list is an array of lists, each containing the neighbors of one of the vertices (out-neighbors if the graph is directed).

Example:





① The overall space required for an adjacency list is $O(n+m)$ { Some books write $O(V+E)$ or $O(|V|+|E|)$ }

n → # of vertices

m → # of edges

② The standard way to implement the adjacency list is using a array (list) of single-linked lists, where the head of the linked list is the vertex and all the connected linked lists are the vertices to which it is connected.

③ We can find neighbors of a vertex i in $\Theta(d_i)$ time, where d_i is the degree of vertex i.

Comparison : [Basic operations on graphs]

	Adjacency list	Adjacency matrix
① Space	$\Theta(V+E)$	$\Theta(V^2)$
② Test if $uv \in E$	$O(1+\min\{\deg(u), \deg(v)\})$ $= O(V)$ (worst case)	$O(1)$
③ Test if $(u,v) \in E$ ↓ directed edge	$O(1+\deg(u)) = O(V)$	$O(1)$
④ List v's (out)-neighbors	$\Theta(1+\deg(v)) = O(V)$	$\Theta(V)$
⑤ List all edges	$\Theta(V+E)$	$\Theta(V^2)$

etc

- - - -

Q: Consider an undirected connected graph with n vertices and no parallel edges.

What are the minimum and maximum numbers of edges respectively that the graph could have?

Sparse vs Dense Graphs

A graph is sparse if the number of edges relatively close to linear in the number of vertices.

Ex: $m = O(n)$ or $m = O(n \log n)$

A graph is dense if the number of edges relatively close to quadratic in the number of vertices.

Ex $m = \Theta(n^2)$

Adjacency Lists (We use this representation in this course)

if a graph is sparse, then most of entries in adjacency matrix are zero (which is a waste of space). Hence, we generally use adjacency list representation for sparse graphs.

* If G has n vertices & m edges then Adjacency list representation requires $\Theta(m+n)$ space.

Comparing the Representations

Q: Which representation is better? Adjacency matrix or Adjacency list.

Remark:

In the rest of the course, Unless explicitly Stated otherwise, all time bounds for graph algorithms assume that the input graph is represented by a adjacency_list.